

| Unit   | Core Modules & Topics                    | Primary Assessment Focus                         | Weightage   |
|--------|------------------------------------------|--------------------------------------------------|-------------|
| Unit 1 | Fundamentals, ER Model & Relational Keys | Architecture, ER Diagrams, Attribute Types, Keys | 15–20 Marks |
| Unit 2 | SQL Syntax, Aggregates & Querying        | DDL/DML/DQL, Operators, Aggregates, Grouping     | 15–20 Marks |
| Unit 3 | Relational Joins & Normalization         | Join Variants, Functional Dependencies, 1NF–BCNF | 15–20 Marks |

### Unit 1: Database Fundamentals and ER Model

5 Core Questions

1. Explain the client-server architecture of a Database Management System (DBMS) with a suitable diagram.
2. Explain the Entity-Relationship (ER) model in detail. Discuss entities, attributes, relationships, cardinality, and participation constraints with suitable examples.
3. Explain the different types of attributes in an ER model with suitable examples.
4. What is a relational database? Explain the characteristics and structure of the relational database model.
5. Explain the different types of keys used in a relational database with suitable examples.

### Unit 2: SQL and Relational Database

7 Core Questions

1. Explain the different categories of SQL commands: DDL, DML, DCL, TCL, and DQL, with suitable examples.
2. Given a table structure and a set of attributes, write SQL statements to create the table and insert the specified records.
3. Explain the different operators used in SQL with suitable examples.
4. Explain the aggregate functions in SQL. Discuss COUNT(), SUM(), AVG(), MIN(), and MAX() with examples.
5. Explain the GROUP BY and HAVING clauses in SQL with suitable examples.
6. Write SQL queries using WHERE, ORDER BY, LIKE, IN, and BETWEEN operators.
7. Write SQL queries to perform INSERT, UPDATE, DELETE, and SELECT operations on a relational table.

### Unit 3: Joins and Normalization

7 Core Questions

1. Explain the concept of joins in SQL. Discuss the different types of joins with suitable examples.
2. Explain Inner Join, Left Outer Join, Right Outer Join, Full Outer Join, Self Join, and Cross Join with suitable examples.
3. What is normalization? Explain the need and objectives of normalization in relational database design.
4. Explain First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) with suitable examples.
5. Explain Boyce-Codd Normal Form (BCNF) and compare it with 3NF.
6. Explain functional dependencies and their role in database normalization.
7. Discuss insertion, deletion, and update anomalies with suitable examples.

**Unit 4: Transactions and Concurrency Control****9 Core Questions**

1. What is a database transaction? Explain the ACID properties of transactions in detail.
2. Explain concurrency control in DBMS and discuss its importance.
3. Explain the problems that may occur due to concurrent execution of transactions.
4. Explain serial and concurrent schedules with suitable examples.
5. What is serializability? Explain conflict serializability and view serializability.
6. Explain the Two-Phase Locking (2PL) protocol in detail with a suitable example.
7. Explain the growing phase and shrinking phase of the Two-Phase Locking protocol.
8. Differentiate between shared locks and exclusive locks.
9. Explain basic 2PL, strict 2PL, and rigorous 2PL.

**Unit 5: Database Recovery****7 Core Questions**

1. What is database recovery? Explain the need for recovery in a DBMS.
2. Explain the different types of failures that can occur in a database system.
3. Explain log-based recovery techniques in detail.
4. Explain deferred database modification and immediate database modification techniques.
5. Explain the concepts of UNDO and REDO operations in database recovery.
6. What is a checkpoint? Explain its role in database recovery.
7. Explain the shadow paging technique for database recovery.

**★ MOST IMPORTANT QUESTIONS (CORE EXAM HOTSPOTS)****High Priority Revision**

1. Explain the ER model in detail with a suitable example.
2. Explain DDL, DML, DCL, TCL, and DQL commands with examples.
3. Write SQL statements to create tables and insert records based on a given table structure.
4. Explain SQL operators and aggregate functions with examples.
5. Explain different types of keys in DBMS.
6. Explain different types of joins with suitable examples.
7. Discuss normalization and explain 1NF, 2NF, 3NF, and BCNF.
8. Explain the ACID properties of database transactions.
9. Explain concurrency control and problems associated with concurrent transactions.
10. Explain the Two-Phase Locking (2PL) protocol in detail.
11. Explain serializability and different types of schedules.
12. Explain database recovery techniques, including UNDO, REDO, and checkpoints.
13. Practice SQL queries involving SELECT, WHERE, GROUP BY, HAVING, ORDER BY, aggregate functions, and joins.